

SAT-based Course Scheduling using Two Encodings

1. Introduction

In this assignment, the course scheduling problem is modeled as a SAT problem.

Given a set of courses, each course must be scheduled in a room for a consecutive number of days within a specified time window. The objective is to determine whether a valid schedule exists that satisfies all constraints.

Each course has:

- preferred start day s_i
- deadline d_i
- duration t_i

Constraints:

1. A course must run for t_i consecutive days.
2. A course must be conducted in a single room.
3. Two courses cannot occupy the same room at overlapping times.

Two different SAT encodings are implemented and compared.

2. Option-1 Encoding

Variables

$z_{i,j,t}$

Meaning: Course i starts on day t in room j

Constraints

Constraint 1: Each course starts at least once

$$\forall (j,t) \ z(i,j,t)$$

This ensures every course is scheduled.

Constraint 2: Each course starts at most once

$$z(i,j,t) \rightarrow \neg z(i,j',t')$$

for all different start assignments.

Constraint 3: No overlapping courses in the same room

If two courses overlap:

$$z(i,j,t) \rightarrow \neg z(i',j,u)$$

This prevents two courses from using the same room at overlapping times.

SAT DIMACS Format

The SAT instance is written in DIMACS format:

p cnf <variables> <clauses>

3. Option-2 Encoding

Variables

Room assignment:

$$x_{i,j}$$

Meaning: course i is assigned to room j

Start time:

$$y_{i,t}$$

Meaning: course i starts on day t

Constraints

Constraint 1: At least one room assigned

$$\forall j \ x(i,j)$$

Constraint 2: At most one room assigned

$$x(i,j) \rightarrow \neg x(i,j')$$

Constraint 3: At least one start time

$$\forall t \ y(i,t)$$

Constraint 4: At most one start time

$$y(i,t) \rightarrow \neg y(i,t')$$

Constraint 5: Prevent overlapping courses in same room

$$\neg(x(i,j) \wedge y(i,t) \wedge x(i',j) \wedge y(i',u))$$

This ensures two overlapping courses cannot be scheduled in the same room.

4. Encoding Statistics

Encoding	Variables	Clauses	2-literal	3-literal	3+-literal
Option-1	125	3248	3245	0	3
Option-2	40	607	131	0	476

5. Performance Comparison

Average results over **100 test cases**:

Solver	Encoding	Avg Time (s)	Avg Memory (KB)
Z3	Encoding-1	0.018785	2.92
Z3	Encoding-2	0.015195	0.28
OR-Tools CP-SAT	Encoding-1	0.013748	266.56
OR-Tools CP-SAT	Encoding-2	0.013042	-8.16
OR-Tools SAT Config	Encoding-1	0.013363	6.96
OR-Tools SAT Config	Encoding-2	0.013240	-6.96

6. Analysis

The results show that Encoding-2 is more efficient than Encoding-1.

1. Encoding-2 reduces variables from 125 to 40.
2. Clauses decrease from 3248 to 607.
3. Encoding-2 consistently produces slightly lower computation times.
4. Due to the smaller SAT formula.

7. Conclusion

Two SAT encodings were implemented for the course scheduling problem.

Experimental results show that Encoding-2 is more efficient, as it reduces the number of variables and clauses while maintaining correctness. Across the tested SAT solvers, OR-Tools solvers demonstrated the best runtime performance, while Z3 also performed efficiently.

Therefore, Encoding-2 combined with modern SAT solvers provides an effective approach for solving the course scheduling problem.